

SmartClass AI: Integrated Classroom Management and Early Warning System with Artificial Intelligence and Automation Support — Design and Preliminary Evaluation

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Abstract—The digital transformation in educational institutions increases teachers' administrative burden while making individual student tracking more difficult. This study presents the architectural design and preliminary feasibility evaluation of the "SmartClass AI" system, developed within the Design Science Research (DSR) framework. The platform comprises seven modules: classroom management, assignment-attendance-grade tracking, real-time messaging, multi-factor risk scoring, large language model (LLM)-assisted intervention plan generation, context-enriched student assistant, and 13 independent n8n automation workflows. In an eight-week pilot study with two teachers and 15 students, the system achieved a Cohen $\kappa = 0.79$ inter-rater agreement coefficient for risk classification and a 4.2/5.0 Likert score for intervention plan evaluation. These findings provide preliminary feasibility evidence regarding operational applicability, making no claims of causality or generalizability, and requiring validation through studies with larger samples and control groups.

Keywords—educational technology; early warning system; large language model; workflow automation; classroom management; Design Science Research; student risk scoring.

I. INTRODUCTION

Artificial intelligence applications in education have gained momentum over the past decade; machine learning, natural language processing, and big data analytics technologies have stepped in where traditional learning management systems (LMS) fall short. [1] [2] Dependence on commercial platforms in resource-constrained environments continues to be one of the structural barriers to digital transformation. [3]

Predicting student achievement and early detection of at-risk students constitutes the core research agenda of the educational analytics field. Romero and Ventura [4] documented the process of educational data mining (EDM) becoming an independent discipline. Baker [5] demonstrated the contribution of EDM methods applied to LMS log data to student failure prediction.

AI-assisted early warning systems (EWS) offer proactive intervention opportunities by monitoring signals such as absenteeism, late submissions, and grade decline. [6] [7] Arnold and Pistilli [8] produced the pioneering empirical evidence in this field with Course Signals; however, this system and similar ones provide risk visualization without automated intervention plan generation. This gap constitutes SmartClass AI's primary point of differentiation. The integration of LLMs into educational systems has expanded rapidly over the past two years. [9] [10] Workflow automation platforms have documented effectiveness in standardizing administrative processes. [11]

Original contributions:

(C1) DSR-based architectural design combining classroom management, early warning, LLM assistant, and automation layers in a single low-license-cost platform supported by open-source components;

(C2) Internally consistent weighted-sum risk scoring algorithm and automated intervention plan generation mechanism;

(C3) Pedagogically constrained LLM assistant based on context-enriched prompting with real-time Airtable data;

(C4) Design and preliminary feasibility evaluation of the zero-touch operational layer comprising 13 n8n automation workflows.

II. RELATED WORK

A. Educational Analytics and EDM

Siemens and Long [12] demonstrated that pedagogical decisions can be improved through systematic analysis of student behavioral data. Campbell, DeBlois, and Oblinger [13] examined the integration of academic analytics into institutional management; Papamitsiou and Economides [6] systematically synthesized empirical EDM evidence.

Credé, Roch, and Kieszczynka [14] calculated the correlation between attendance behavior and grade point average at $r = 0.41$ via meta-analysis, a finding that justifies positioning absenteeism as the primary variable in risk models. Romer [15] documented the relationship between assignment submission behavior and academic achievement.

B. Early Warning Systems

Classical methods such as logistic regression and decision trees have been prominent in the EWS literature. [8] [16] More recent research has adapted LSTM and Transformer architectures to educational environments, reporting F1 scores above 90% with large and balanced datasets. [17] [18] However, high accuracy rates generally require large datasets, which poses a distinct challenge for institutions operating under limited data conditions. [19]

Essa and Ayad [20] showed that ensemble approaches produce more reliable risk scores compared to single models. A common shortcoming of existing EWS solutions is their limitation to detecting risk signals without generating automated pedagogical intervention steps; filling this gap constitutes the primary motivation of this work.

C. LLMs and Context-Enriched Prompting

The most significant problem with LLMs in educational contexts is the risk of generating hallucinated content. [9] To mitigate this risk, context-enriched prompting techniques provide the model with domain-specific data and constraints. [21] Lewis et al. [22] proposed generation architectures leveraging external document repositories; Gao et al. [23] mapped the broad application spectrum of such approaches in knowledge-intensive domains. In SmartClass AI, real-time Airtable data is injected into the prompt context; however, vector indexing or semantic search is not included.

D. Low-Code Automation Platforms

Visual automation tools such as n8n, Zapier, and Make have documented effectiveness in reducing administrative workload. Richardson [11] reported that these tools provide significant time savings in routine notification and reporting tasks. Trueman and Hartley [24] drew attention to the relationship between time management and academic performance.

III. RESEARCH METHODOLOGY

This study was conducted within the Design Science Research (DSR) framework as defined by Hevner et al. [25]. DSR is a research paradigm that designs artificial artifacts to produce solutions for human and organizational problems and rigorously evaluates them. The study covers three core DSR outputs: (i) problem identification and motivation, (ii) design and development, (iii) prototype preliminary evaluation.

A. Study Type and Scope

This study is a System Design and Preliminary Feasibility Evaluation study. It is not a controlled experimental study; the findings provide preliminary evidence regarding the system's operational applicability and make no claims of causality. The long-term impact on student academic

achievement is a subject for future studies with control groups and larger samples.

B. Participants

The preliminary evaluation was conducted with two teachers and 15 undergraduate students who participated on a voluntary basis throughout the 2025–2026 fall semester. Informed consent was obtained from all participants, data were anonymized, and used solely for system evaluation purposes. Context data transmitted to the LLM was structured to exclude directly identifying information.

C. Data Collection

Instruments used in the eight-week evaluation: (i) system logs: 120 attendance records, 45 assignment submissions, 210 grade entries, 38 messaging sessions; (ii) risk classification comparison: system vs. two independent teacher assessments; (iii) intervention plan evaluation form: 5-point Likert scale (14 forms) based on pedagogical appropriateness, feasibility, and neutral language criteria; (iv) technical performance log: webhook response times and automation execution records.

IV. SYSTEM ARCHITECTURE

SmartClass AI has a five-layer cloud-native architecture structured on the principle of Separation of Concerns. [26] Each layer is designed to be independently scalable and replaceable.

Figure 1. SmartClass AI — Layered System Architecture

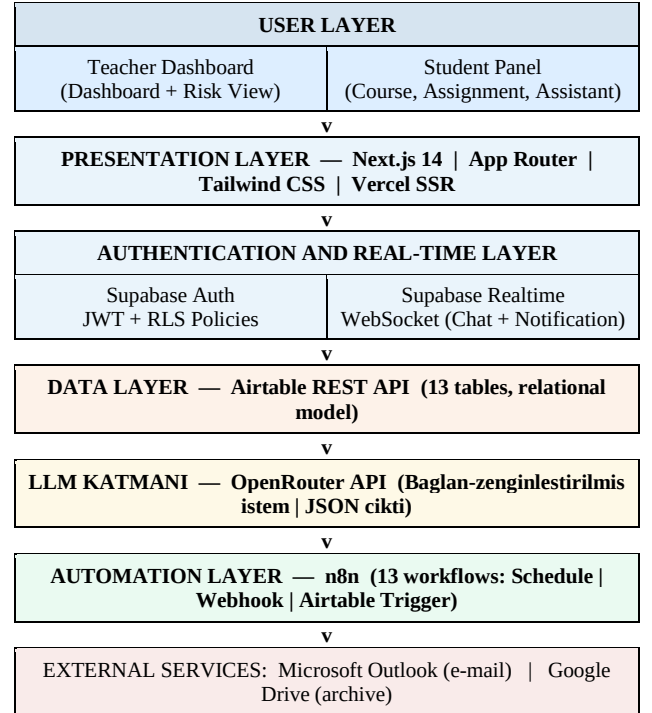


Table I. System Layers, Technologies, and Responsibilities

Layer	Technology	Responsibility
Presentation	Next.js 14 + Vercel	SSR, routing, UI/UX
Auth	Supabase Auth (JWT)	Authentication, RLS, roles
Real-Time	Supabase Realtime	Chat and notification WS
Data	Airtable REST API	Relational data model
LLM	OpenRouter API	Analysis, assistant, plan
Automation	n8n (13 workflows)	Trigger + action

A. Authentication and Data Layer

Supabase Auth provides JWT-based session management and Row-Level Security (RLS) policies. Core Airtable tables: Users, Classes, Class_Memberships, Assignments, Assignment_Submissions, Attendance, Grades, Risk_Summaries, AI_Predictions, Risk_Signals, Intervention_Plans, Notifications, and Weekly_Reports. Workflow 09 automatically performs data quality auditing on a weekly basis. [27]

B. LLM Integration

Before each LLM call, student, class, and risk data retrieved in real time from Airtable are injected into the prompt context. This context-enriched prompting approach does not include vector indexing, embeddings, or semantic search. Per the pedagogical constraint policy, the model operates with explicit system instructions that prevent generation of exam answers, identifiable personal data, and confidential system information.

V. FUNCTIONAL MODULES

A. Classroom and Attendance Management

The teacher creates a class with a unique enrollment code; the student sends an enrollment request using this code. Workflow 04 notifies the teacher in real time; after approval, the student gains access to all class content. Attendance records are kept on a session basis (Present/Absent/Late), and Workflow 03 automatically sends alerts to students who exceed the absenteeism threshold and to the teacher. [28]

B. Assignment, Grade, and Notification Module

Assignments are defined with Easy/Medium/Hard difficulty levels; Workflow 02 automatically sends reminders as the deadline approaches. Grades are recorded cumulatively, and Workflow 05 automatically notifies students of new grade announcements. The Supabase Realtime WebSocket infrastructure supports class group chat and teacher-student private messaging, while Workflow 13 automatically archives assignment files to Google Drive. [29]

C. Early Warning and Risk Algorithm

The early warning engine calculates a risk score (RS) on a 0–100 scale for each student using the following weighted-sum formula:

$$RS = 100 \times [0.35 \times A + 0.30 \times (1 - S) + 0.35 \times (1 - N')]$$

Variable definitions: A, S, and N' are all normalized variables in the [0,1] range. A = absenteeism rate (absent sessions / total sessions); S = assignment submission rate (submitted / total assignments); N' = composite grade score, calculated as follows:

$$N' = 0.70 \times N_{avg} + 0.30 \times Trend_{norm}$$

where N_avg = normalized overall grade average; Trend_norm = slope coefficient (beta) obtained from least-squares linear regression applied to the last three grades, compressed into the [0,1] range via the following min-max transformation:

$$Trend_{norm} = clip((beta - beta_{min}) / (beta_{max} - beta_{min}), 0, 1)$$

where beta_min = -10 and beta_max = +10 points/session are fixed; values outside this range are clipped to 0 or 1. Positive slope (grade increase) → Trend_norm → 1; negative slope (grade decline) → Trend_norm → 0.

Weight values were determined heuristically based on Credé et al.'s [14] absenteeism–GPA meta-analysis (r = 0.41) and Romer's [15] findings on assignment submission and performance. Empirical optimization of these weights is deferred to a future data collection phase.

Risk categories: Low (RS ≤ 40), Medium (41–60), High (61–84), Critical (RS ≥ 85). These thresholds are operational preferences established in design sessions with teachers and require statistical validation. For students in the High/Critical category, Workflow 07 generates an automated intervention plan, and Workflow 10 sends a risk signal e-mail to the teacher.

Figure 2. Sample LLM Intervention Plan Output (Workflow 07 — Anonymized)

[RISK SIGNAL REPORT — HIGH CATEGORY]

Student ID: S-007 | Class: CSC-301 | RS = 78 | Category: High

Triggering Risk Factors:

- Absenteeism rate: 42% (threshold: 30%)
- Late submissions: 3 / 5 assignments
- Grade trend (Trend_norm): 0.18 — strong decline signal

Recommended Intervention Steps:

1. Schedule an individual meeting within this week.
2. Provide extended time and support for missing assignments.
3. Apply intensive attendance monitoring for the next 2 weeks.
4. Coordinate with academic counseling if necessary.

Estimated duration: 14 days | Generated by: n8n Workflow 07 + OpenRouter LLM

D. LLM Assistant and Automation Workflows

LLM integration operates in two contexts: (i) Student assistant: study support and system usage guidance; (ii) Teacher assistant: class summary, risk interpretation, weekly lesson recommendations (Workflow 12). Workflow 11 returns webhook requests from external systems in real time. [9] [30]

Table II. n8n Automation Workflows — Trigger and Function

No	Trigger	Function
01	Weekly Schedule (Mon 09:00)	Class-based risk report generation and recording
02	Daily Schedule	Assignment deadline reminder e-mail
03	Daily Schedule	Warning e-mail when absenteeism threshold is exceeded
04	Airtable Trigger	Teacher notification on enrollment request
05	Airtable Trigger	Student e-mail notification on grade announcement
06	Weekly Schedule	Individual LLM performance prediction generation
07	Airtable Trigger	High/Critical risk — intervention plan generation
08	Weekly Schedule	Weekly progress summary e-mail to student
09	Weekly Schedule	Data quality audit and audit record
10	Weekly Schedule	Risk signal detection and teacher alert
11	Webhook (HTTP POST)	Real-time LLM assistant response (API)
12	Weekly Schedule	LLM-assisted weekly lesson and class recommendation
13	Airtable Trigger	Archiving assignment files to Google Drive

VI. PRELIMINARY EVALUATION

A. Evaluation Setup and Limitations

This section presents the findings of the controlled pilot study conducted within the DSR framework. No independent educational outcome was measured in the study; evaluation focused on operational consistency, risk classification reliability, and perceived quality of intervention plans. There is no control group and no baseline comparison; this positions the study at the level of preliminary feasibility evidence.

B. Findings

Risk classification reliability was calculated as Cohen κ = 0.79 between the system output and the assessments of two

independent teachers. [31] According to the Landis and Koch [31] scale, this value corresponds to the ‘strong agreement’ category. This measurement gauges the consistency between the system and teacher assessments; it does not evaluate prediction accuracy against an independent ground truth.

The 14 intervention plans generated by the LLM received an average Likert score of 4.2 ± 0.6 . Evaluation dimensions: pedagogical appropriateness, feasibility, and absence of student-blaming language. Automation workflows completed 88 of 91 planned executions over eight weeks, achieving 96.7% operational consistency.

Table III. Summary of Preliminary Evaluation Findings

Measurement	Result
Risk classif. inter-rater agreement (κ)	0.79 (Strong)
Intervention plan Likert score	4.2 ± 0.6 / 5.0
Automation operational consistency	88/91 = 96.7%
Webhook average response time	1.4 seconds
Detected risk signals	7 (8-week period)
Independent educational outcome	— (future study)
Baseline comparison	— (future study)

VII. DISCUSSION

Preliminary findings provide early evidence regarding SmartClass AI’s operational applicability in risk classification and pedagogical intervention plan generation. Compared with Course Signals [8] and similar EWS solutions, the system’s primary difference is proposing automated and structured intervention steps beyond risk visualization. However, the long-term pedagogical impact of this difference cannot be established without validation through a study with a control group and a larger sample.

The current design has the potential to offer a more accessible structure compared to similar platforms in resource-constrained educational environments, thanks to open-source components and low licensing costs. However, this claim cannot be asserted as a definitive conclusion without systematic cost comparisons and scalability experiments.

Key limitations of the pilot study: (i) two teachers and 15 students are insufficient for generalizability; (ii) there is no control group; (iii) risk weights have not been statistically optimized; (iv) intervention plan satisfaction is not an independent pedagogical assessment. Airtable REST API rate limits (5 requests/second on the free tier) may create bottlenecks in large-scale institutional deployments;

replacement with a PostgreSQL-based infrastructure is recommended in such cases. [19] [32]

VIII. RESEARCH RECOMMENDATIONS

In light of the preliminary feasibility findings and identified limitations, four priority research directions are recommended:

A. Large Sample, Control Group, and Educational Outcomes

A study with experimental and control groups, at least 200 students, and spanning a full academic term should be designed to measure the system's true pedagogical impact. Independent outcome variables (grade improvement, dropout rate, change in assignment submission rate) should be measured, and a baseline condition using only manual assessment should be included. [6] [19]

B. Risk Weight and Threshold Optimization

The current heuristic weights need to be empirically optimized using logistic regression or k-fold cross-validation with actual achievement and dropout data. The context-dependence of beta boundary values (beta_min, beta_max) should also be examined separately. [17] [18]

C. LLM Output Quality and Bias Auditing

Systematic examination of LLM-generated intervention plans through independent expert evaluation along dimensions of pedagogical accuracy, bias, and safety is of critical importance. [9] [10]

D. Legal Compliance and Privacy Architecture

The integration into the system of a KVKK/GDPR compliance architecture comprising data minimization, anonymization, consent management, and audit trail components constitutes a priority research agenda. [33]

IX. CONCLUSION

This study presented the architectural design, functional modules, and preliminary feasibility evaluation findings of the SmartClass AI platform developed within the DSR framework. The internally consistent risk scoring formula ($RS = 100 \times [\dots]$), min-max normalized Trend_norm calculation, context-enriched LLM integration, and operational layer comprising 13 automation workflows demonstrate that the system has completed its initial feasibility step as an integrated educational technology solution.

The Cohen $\kappa = 0.79$ inter-rater agreement and 4.2/5.0 intervention plan satisfaction are interpreted as early indicators. These findings require validation through large-sample studies in which the system is evaluated against independent educational outcomes and a comparative baseline. This study is positioned as the starting point and prototype reference for that research process.

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